Power, mains & fields

Use energy equations

1 Energy & power

$P = VI$; $P = I^2R$; $E = Pt$; $E = QV$. Use W, V, A, Ω, J, s, C.

12 V × 2 A = 24 W; for 30 s → 720 J. Double I at fixed R → four times heating power.

2 Mains & protection

UK: **230 V, 50 Hz ac;** polarity reverses. Cell: dc.

Live brown; neutral blue near 0 V; earth green/yellow, 0 V. Earth normally has no current.

Live-to-case fault → earth current → fuse/breaker disconnects. Live can remain dangerous with switch open.

Transmit safely

3 National Grid

Step up V for transmission; for same P = VI, current falls.

Cable heating **I^2R** falls. Step down V for consumers. High voltage does not lower cable resistance.

4 Charging by friction

Electrons transfer between insulators; protons remain in nuclei. Equal opposite charges result.

Gain electrons → negative. Lose electrons → positive. Like charges repel, unlike attract.

Explain static

5 Electric field

Positive sphere: field points out

Force on a positive test charge defines direction. Isolated + sphere: outwards; − sphere: inwards.

Closer to sphere → stronger field. Lines never cross.

6 Sparks

Large charge imbalance can give a strong field. Air ionises and becomes conducting.

Charge flows suddenly through the air, reducing the imbalance. Force can act before contact.